ACIDS AND BASES 1

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Acids

The definitions of acids and bases have changed over the years. One way of classifying substances was by taste: sweet, sour, salt or bitter. After a while it was noticed that the sour substances had many other properties in common; e.g. they changed the colour of blue litmus to red and corroded some metals. The Latin verb *acere* means 'to make sour'. This gave the term **acid**. Some common, naturally-occurring acids that have been known for a long time include ethanoic acid (acetic acid or vinegar, CH_3COOH), citric acid (from citrus fruits, 2-hydroxypropane-1, 2, 3-tricarboxylic acid, see figure 1) and lactic acid (from sour milk, $CH_3CH(OH)COOH$).

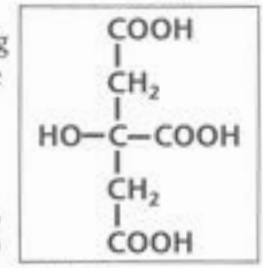


Figure 1

1 What is the general term for substances such as litmus that change colour in the presence of acids and bases? Why do you think they are called this?

2 Give the molecular formula for citric acid. (See figure 1.)

In 1661, Robert Boyle (1627–1691) published a work called *The Sceptical Chymist* in which, among other things, he distinguished between acids and bases (although he referred to bases as alkalis). He noted that all acids turned a particular vegetable indicator from blue to red and all alkalis turned the indicator green. Boyle found that some substances did not change the colour of this indicator and he concluded that these substances were neutral. Around this time it was also proposed that salts are formed when acids react with bases.

Antoine Laurent Lavoisier (1743–1794) was one of the first people to investigate what was responsible for giving an acid its properties. By 1776 he had come to the conclusion that one element was present in all of the acids he knew; hence this element must cause an acid to behave as it does. This element was **oxygen**, named from the Greek terms *oxein* (sour) and *gennen* (to generate). Lavoisier was wrong, because he only knew the formulae of a few acids.

Around 1815, Sir Humphry Davy (1778–1829) came to the conclusion that acidity was not caused by oxygen because he had discovered that there were acids that did not contain any oxygen. Acidity must be caused by something else.

3 Give the names and sources of three naturally-occurring acids.

Davy showed that muriatic acid (now called hydrochloric acid) could be reacted with mercury to give only mercury chloride and hydrogen as products, no oxygen.

4 Balance this equation that shows Davy's experiment: $HCl + Hg \rightarrow Hg_2Cl_2 + H_2$

Davy suggested that acidity was caused by hydrogen, because this element was common to all acids.

5 If all acids contain hydrogen, does this mean that all compounds containing hydrogen are acids? Explain your answer.

Around 1840, Baron Justus von Liebig (1803–1873) came up with the first adequate definition of an acid: *A compound containing hydrogen in a form in which it can be replaced by a metal*. This classification is still very useful today.

Bases

Bases were thought of as simply compounds that could neutralise acids. This was until Svante August Arrhenius (1859–1927), came up with a more complex theory in 1887. He concluded that acids are substances that, when dissolved in water, broke up (dissociated) to give hydrogen (H^+) ions. In other words, substances that dissolve in water (dissociate) to produce hydrogen ions are called acids. Bases are substances that produce hydroxide (OH^-) ions when dissolved in water. This is known as the **Arrhenius concept**. Note that the hydrogen ion does not exist by itself in water. It is hydrated and is often written as H_3O^+, the hydronium ion.

6 Robert Boyle classified substances into three groups. What are these groups?

7 Which chemist believed that oxygen was the element responsible for a compound being an acid?

8 Which chemist is credited with associating hydrogen with acidic properties?

9 *A substance that can neutralise an acid producing a salt plus water.* What does this define?

10 Give the formula for the hydrogen ion.

11 Give the formula for the hydroxide ion.

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Dissociation

12 The following reactions show how some substances behave when they dissolve in water. Copy each equation, name the substances present and state whether the substances are acids or bases according to Arrhenius.

EXAMPLE: $HCl\ (aq) + H_2O\ (l) \rightarrow H_3O^+\ (aq) + Cl^-\ (aq)$
Hydrogen chloride (hydrochloric acid) + water → hydrogen ions + chloride ions (acid)

a $HNO_3\ (aq) + H_2O\ (l) \rightarrow H_3O^+\ (aq) + NO_3^-\ (aq)$

b $NaOH\ (s) + H_2O\ (l) \rightarrow Na^+\ (aq) + OH^-\ (aq)$

c $CH_3COOH\ (l) \rightarrow CH_3COO^-\ (aq) + H_3O^+\ (aq)$

d $NH_4OH\ (aq) + H_2O\ (l) \rightarrow NH_4^+\ (aq) + OH^-\ (aq)$

Brønsted–Lowry The Arrhenius concept of acids and bases had some shortcomings:

- Only aqueous solutions were considered
- Only substances containing hydroxides could be bases.

In 1923, two chemists separately and independently came up with broader definitions of acids and bases. Johannes Brønsted (1879–1947) and Thomas Lowry (1874–1936) focused on the role of protons (hydrogen ions). Their definitions:

- Acids are substances from which protons can be removed
- Bases are substances that bind protons.

These are often quoted as:

- Acids donate protons
- Bases accept protons.

13 The following reactions show how some acids and bases react. Copy each equation, name the substances present and decide which species are acids and bases according to Brønsted–Lowry.

EXAMPLE: $HCl\ (aq) + NH_3\ (aq) \rightarrow NH_4^+\ (aq) + Cl^-\ (aq)$
Hydrochloric acid (HCl) has transferred a proton to ammonia (NH_3) forming ammonium and chloride ions. HCl is the acid and NH_3 is the base.

a $NH_3\ (aq) + H_2O\ (l) \rightarrow NH_4^+\ (aq) + OH^-\ (aq)$

b $HNO_3\ (l) + H_2O\ (l) \rightarrow NO_3^-\ (aq) + H_3O^+\ (aq)$

c $NaOH\ (s) + H_2O\ (l) \rightarrow Na^+\ (aq) + OH^-\ (aq)$

Note that all Arrhenius acids are acids according to the Brønsted–Lowry definitions, but not all Brønsted–Lowry acids are Arrhenius acids.

14 Copy each statement and decide whether it applies to acids or bases or both.

a Have a bitter taste

b Have a sour taste

c Aqueous solutions are electrolytes (conduct electricity)

d Have a slippery feel

e Turn blue litmus red

f Turn red litmus blue

g React with many metals to form hydrogen and a salt

Research **15** An American chemist named Gilbert Newton Lewis (1875–1946) came up with yet another definition of acids and bases. Summarise his definitions of acids and bases.

16 When an acid and a base react, a salt is formed. Sulfuric acid forms sulfates, nitric acid forms nitrates, and so on. Copy and complete the table. The first one has been done for you.

Acid	Formula	Salts formed
Hydrochloric	HCl	Chlorides
	HBr	
	HNO_3	
Sulfuric		
		Ethanoates (acetates)

17 Define *salt*.

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